SESSION 14 – Chi-Square Test

Task 1. Create a contingency table

Question: Create a contingency table in Excel or Google Sheets showing the relationship between users' favorite food delivery app (Zomato, Swiggy, Uber Eats) and their preferred payment method (Paytm, PhonePe, Credit Card), using at least 20 responses.

Answer:

The following is a sample dataset of 30 users.

User	Favorite App	Payment Method
1	Zomato	PhonePe
2	Swiggy	Paytm
3	Zomato	Credit Card
4	Uber Eats	PhonePe
5	Swiggy	PhonePe
6	Zomato	Paytm
7	Uber Eats	Credit Card
8	Swiggy	PhonePe
9	Zomato	PhonePe
10	Swiggy	Credit Card
11	Zomato	PhonePe
12	Uber Eats	Paytm
13	Swiggy	Paytm
14	Zomato	Credit Card
15	Uber Eats	PhonePe
16	Zomato	Paytm
17	Swiggy	PhonePe
18	Zomato	PhonePe
19	Uber Eats	Credit Card
20	Swiggy	Paytm
21	Zomato	PhonePe
22	Swiggy	Credit Card
23	Uber Eats	PhonePe
24	Zomato	Paytm
25	Swiggy	PhonePe

26	Zomato	Credit Card
27	Uber Eats	Paytm
28	Swiggy	PhonePe
29	Zomato	PhonePe
30	Swiggy	Paytm

Contingency table

Favorite App	Paytm	PhonePe	Credit Card	Total
Zomato	3	5	3	11
Swiggy	4	5	2	11
Uber Eats	2	2	2	6
Total	9	12	7	28

Correction: The sample above contains 30 responses. The correct row and column totals are shown below for use in the remaining tasks.

Correct contingency table for Task 1

After counting all 30 responses, use this corrected table for your calculations:

Favorite App	Paytm	PhonePe	Credit Card	Total
Zomato	3	6	3	12
Swiggy	4	5	2	11
Uber Eats	2	3	2	7
Total	9	14	7	30

How to create it in Excel

- Enter the 30 responses in three columns: User, Favorite App, Payment Method.
- Select the data.
- Click Insert → PivotTable.
- Drag Favorite App into Rows.
- Drag Payment Method into Columns.
- Drag Payment Method into Values and make sure it uses Count.

This creates the contingency table automatically.

Task 2. Calculate expected counts

Question: Given the observed counts of music streaming preferences among males and females, calculate the expected counts for each cell assuming independence.

Answer:

Observed frequency table

Music App	Males	Females	Total
Spotify	12	18	30
Gaana	8	7	15
YouTube Music	10	15	25
Total	30	40	70

Formula:

E=Row Total×Column TotalGrand TotalE = \frac{\text{Row Total}\times\text{Column Total}}{\text{Grand Total}}E=Grand TotalRow Total×Column Total

Calculations

Spotify – Males

E=30×3070=12.86E = \frac{30\times30}{70}=12.86E=7030×30=12.86

Spotify – Females

E=30×4070=17.14E = \frac{30\times40}{70}=17.14E=7030×40=17.14

Gaana – Males

E=15×3070=6.43E = \frac{15\times30}{70}=6.43E=7015×30=6.43

Gaana – Females

E=15×4070=8.57E = \frac{15\times40}{70}=8.57E=7015×40=8.57

YouTube Music – Males

E=25×3070=10.71E = \frac{25\times30}{70}=10.71E=7025×30=10.71

YouTube Music – Females

E=25×4070=14.29E = \frac{25\times40}{70}=14.29E=7025×40=14.29

Expected frequency table

Music App	Males	Females
Spotify	12.86	17.14
Gaana	6.43	8.57
YouTube Music	10.71	14.29

Conclusion: These are the expected counts if gender and music streaming preference are independent.

Task 3. Chi-square test of independence

Question: Perform a chi-square test of independence using the contingency table from Task 1 to determine if there is a significant association between app choice and payment method. Show calculation steps and state your conclusion.

Answer:

We want to check whether the favorite food delivery app and preferred payment method are associated.

Step 1: State the hypotheses

- Null Hypothesis (H_0): Favorite food delivery app and payment method are independent. There is no significant association.
- Alternative Hypothesis (H_1): Favorite food delivery app and payment method are associated.

Step 2: Observed frequency table

App	Paytm	PhonePe	Credit Card	Total
Zomato	3	6	3	12
Swiggy	4	5	2	11
Uber Eats	2	3	2	7
				30
Total	9	14	7	

Step 3: Calculate expected frequencies

Formula:

E=Row Total×Column Total30E = \frac{\text{Row Total}\times\text{Column Total}}{30}E=30Row Total×Column Total

For example, Zomato and Paytm:

E=12×930=3.6E = \frac{12\times9}{30}=3.6E=3012×9=3.6

Expected frequency table

App	Paytm	PhonePe	Credit Card
Zomato	3.60	5.60	2.80
Swiggy	3.30	5.13	2.57

Uber Eats 2.10 3.27 1.63

Step 4: Calculate Chi-square value

Formula:

$\chi^2 = \sum \frac{(O-E)^2}{E}$

After calculating the contributions from all nine cells:

$\chi^2 \approx 0.529$

Degrees of freedom:

$df=(r-1)(c-1)$

$df=(3-1)(3-1)=4$

At a 5% significance level:

- Chi-square value = 0.529
- Degrees of freedom = 4
- P-value ≈ 0.971
- Significance level = 0.05

Step 5: Conclusion

Since the p-value (0.971) is greater than 0.05, we fail to reject the null hypothesis.

There is not enough statistical evidence to conclude that favorite food delivery app and preferred payment method are associated.

Note: This is a small simulated dataset. Several expected cell counts are below 5, so the usual chi-square approximation may not be reliable. A larger sample would be preferable.

Task 4. Chi-square goodness of fit test

Question: For a new Flipkart product launch, suppose you surveyed 30 users and observed the following: 18 prefer the new product and 12 do not. If the company claims that 70% of users will prefer the new product, perform a chi-square goodness of fit test.

Answer:

Step 1: Given information

- Total users = 30
- Users who prefer the product = 18

- Users who do not prefer the product = 12
- Claimed preference = 70%

Step 2: State the hypotheses

- Null Hypothesis (H_0): 70% of users prefer the new product.
- Alternative Hypothesis (H_1): The proportion of users who prefer the new product is different from 70%.

Step 3: Calculate expected frequencies

Expected preference:

E=30×70100=21E=30\times\frac{70}{100}=21E=30×10070=21

Expected non-preference:

E=30×30100=9E=30\times\frac{30}{100}=9E=30×10030=9

Step 4: Observed and expected table

Response	Observed (O)	Expected (E)
Prefer product	18	21
Do not prefer	12	9
Total	30	30

Step 5: Calculate Chi-square value
χ2=∑(O−E)2E\chi^2=\sum\frac{(O-E)^2}{E}χ2=∑E(O−E)2

For users who prefer the product:

(18−21)221=921=0.4286\frac{(18-21)^2}{21}=\frac{9}{21}=0.428621(18−21)2=219=0.4286

For users who do not prefer the product:

(12−9)29=99=1\frac{(12-9)^2}{9}=\frac{9}{9}=19(12−9)2=99=1

Therefore:

χ2=0.4286+1=1.4286\chi^2=0.4286+1=1.4286χ2=0.4286+1=1.4286

Degrees of freedom:

df=2−1=1df=2-1=1df=2−1=1

At a 5% significance level:

- Chi-square value = 1.4286
- Degrees of freedom = 1
- P-value ≈ 0.232
- Significance level = 0.05

Step 6: Conclusion

Since the p-value (0.232) is greater than 0.05, we fail to reject the null hypothesis.

There is not enough statistical evidence to conclude that the actual customer preference differs from the company's claimed 70%.